

Beginning Algebra Test I

Name
Date
Course

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the numbers, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Simplify the following polynomial, write in descending order of powers, and state its degree.

① $3x^5 + x^2 - 2x + 4x^5 - 3 + 2x - 4x^2 - 5x^2 + 10$

Add the following polynomials.

② a) $(7x^2 + 3x + 11) + (8x^2 - 5x - 2)$

b) $(9x^3 - 5x^2 + 8) + (-2x^2 - 4x + 7)$

Subtract the following polynomials.

③ a) $(-6x^3 + 2x^2 - 4x) - (-2x^2 + 9x - 1)$

b) $(4x^3 - 1) - (8x^3 + 5)$

Multiply.

④ a) $3x^2(5x^4 - 3x^2 + 9x - 4)$ b) $-2x(7x^2 + 4x)$

Divide.

⑤ a) $\frac{12x^5 - 4x^3 + 8x^2}{4x}$ b) $\frac{72x^5 - 45x^4 + 63x^2}{-9x^2}$

Expand.

⑥ a) $(5x + 2)(3x - 8)$ b) $(10x + 7)(8x - 3)$

⑦ a) $(-2a + b)(5a + 4b)$ b) $(2x + 1)(8x^2 - 4x - 3)$

⑧ a) $(2x - 4)^2$ b) $(3x + 2y)^2$

Factor the following expressions completely.

⑨ a) $x^2 - 7x + 6$ b) $6x^2 - 13x - 15$

⑩ a) $g^2 + 2gh - 8h^2$ b) $x^2 - 64$

⑪ a) $25x^2 - 81z^2$ b) $-49 + x^2$

⑫ a) $z^2 - 1$ b) $9 - y^2$

⑬ a) $-x^2 + 100$ b) $9 - 25x^2$

⑭ a) $x^4 - x^2 - 6$ b) $x^4 - 5x^2 - 36$

⑮ a) $4x^4 + 11x^2 - 3$ b) $3x^3 - 3x^2 + 2x - 2$

⑯ a) $5x^3 + 2x^2 - 45x - 18$ b) $y^5 + y^4 - 16y - 16$

⑰ $2x^3 - 20x^2 + x - 10$

⑱ $27x^3 - 12x$

⑲ $-x^3 - 5x^2 + 50x$

⑳ $3x^6 - 3x^2$

㉑ $2x^5 - 4x^3 - 16x$

$$\textcircled{22} \quad 8a^3b - 8ab^3$$

$$\textcircled{23} \quad 12ab^3 - 8a^2b^2 + 24ab^2$$

$$\textcircled{24} \quad 18x^6y + 12x^5y - 18x^3y - 12x^2y$$

$$\textcircled{25} \quad 24a^4b^3 - 4a^3b^4 - 8a^2b^5$$

Beginning Algebra Test I Answers

① $7x^5 - 8x^2 + 7, D=5$

② a) $15x^2 - 2x + 9$

b) $9x^3 - 7x^2 - 4x + 15$

③ a) $-6x^3 + 4x^2 - 13x + 1$

b) $-4x^3 - 6$

④ a) $15x^6 - 9x^4 + 27x^3 - 12x^2$

b) $-14x^3 - 8x^2$

⑤ a) $3x^4 - x^2 + 2x$

b) $-8x^3 + 5x^2 - 7$

⑥ a) $15x^2 - 34x - 16$

b) $80x^2 + 26x - 21$

⑦ a) $-10a^2 - 3ab + 4b^2$

b) $16x^3 - 10x - 3$

⑧ a) $4x^2 - 16x + 16$

b) $9x^2 + 12xy + 4y^2$

⑨ a) $(x-6)(x-1)$

b) $(6x+5)(x-3)$

⑩ a) $(g-2h)(g+4h)$

b) $(x-8)(x+8)$

⑪ a) $(5x-9z)(5x+9z)$

b) $(x-7)(x+7)$

⑫ a) $(z-1)(z+1)$

b) $(3-y)(3+y)$

⑬ a) $(10-x)(10+x)$

b) $(3-5x)(3+5x)$

⑭ a) $(x^2-3)(x^2+2)$

b) $(x^2+4)(x-3)(x+3)$

⑮ a) $(2x-1)(2x+1)(x^2+3)$

b) $(3x^2+2)(x-1)$

⑯ a) $(x-3)(x+3)(5x+2)$

b) $(y-2)(y+2)(y^2+4)(y+1)$

⑰ $(2x^2+1)(x-10)$

⑱ $3x(3x-2)(3x+2)$

⑲ $-x(x+10)(x-5)$

⑳ $3x^2(x-1)(x+1)(x^2+1)$

㉑ $2x(x-2)(x+2)(x^2+2)$

㉒ $8ab(a-b)(a+b)$

㉓ $4ab^2(3b-2a+6)$

㉔ $6x^2y(x^3-1)(3x+2)$

㉕ $4a^2b^3(3a-2b)(2a+b)$